

Tuesday Warm-up: Electrostatics

Prof. Jordan C. Hanson

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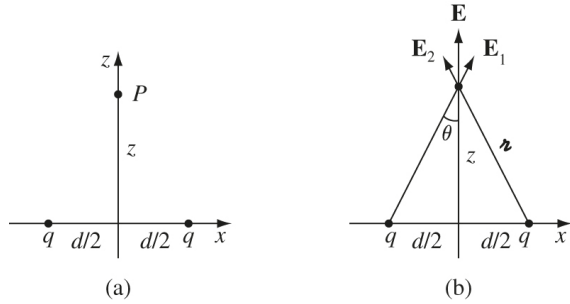


Figure 1: A dipole of two positive charges.

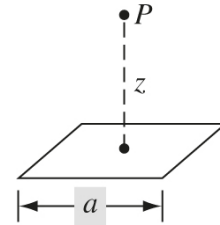


Figure 2: A square of size a in the xy -plane centered at the origin. The point P is a distance z above the origin.

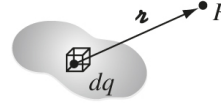
1 Memory Bank

- Permittivity of free space: $\epsilon_0 = 8.85 \times 10^{-12} \text{ N}^{-1} \text{ C}^2 \text{ m}^{-2}$
- Coulomb's Law, discrete

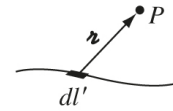
$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i \quad (1)$$

- Coulomb's Law, continuous

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau' \quad (2)$$



(a) Continuous



(b) Line charge, λ

Figure 3: Charge distributions each have a unique dq .

3. Consider Fig. 2. Suppose that, when viewing the square from P , there is a charge $-q$ at the lower right corner, a charge q at the upper right corner, a charge $-q$ at the upper left corner, and a charge q at the lower left corner. Find the electric field at P .

2 Discrete Charge Distributions

1. Consider Fig. 1. Find the electric field a distance z above the midpoint between two equal charges q a distance d apart. Check that the units are N C^{-1} .
2. (a) Using the result from exercise 1, take the following limits: (i) $z \gg d$ (ii) $z \rightarrow 0$. (b) How would the calculation change if the charge at $(d/2)\hat{i}$ was negative?

3 Continuous Charge Distributions

1. Consider Fig. 1. Suppose that instead the q was spread uniformly between $[-d/2, d/2]$, with a density λ (C m^{-1}). Find the electric field a distance z above the midpoint.